

Name: DHRUV GAVANKAR	Roll No.: S007
Class: SYIT	Batch:1
Date of Assignment: 19/08/2026	Date/Time of Submission: 19/08/2026

```
import pandas as pd
```

```
df = pd.read_csv("Students.csv")
```

6a. Write a Python program to import a CSV dataset using Pandas and display the first 5 and last 5 records.

Print top 5 values

```
print(df.head())
```

	Roll No	Name	Department	Marks	Attendance
0	101	Dhruv	BSc IT	78	85
1	102	Aryan	BSc CS	65	90
2	103	Sahil	BSc IT	88	92
3	104	Sumeet	BSc CS	72	80
4	105	Kaif	BSc IT	55	70

Double-click (or enter) to edit

Print last 5 values

```
print(df.tail())
```

	Roll No	Name	Department	Marks	Attendance
2	103	Sahil	BSc IT	88	92
3	104	Sumeet	BSc CS	72	80
4	105	Kaif	BSc IT	55	70
5	106	Usaid	BSc IT	81	75
6	107	Divyanshu	BSc IT	67	87

6b. Write a Python program to analyze the dataset by displaying the number of rows and columns, column names, data types, and basic statistical information.

```
print(df.shape) #number of rows and columns
```

```
(7, 5)
```

```
print(df.columns)
```

```
Index(['Roll No', 'Name', 'Department', 'Marks', 'Attendance'], dtype='object')
```

```
print(df.dtypes)
```

```
Roll No      int64
Name         object
Department   object
Marks        int64
Attendance   int64
dtype: object
```

6c. Write a Python program to filter records from the dataset using conditions. For example, display students having marks greater than 60 or employees having salary greater than 40,000.

```
result = df[df["Attendance"]>=90]
print(result)
```

	Roll No	Name	Department	Marks	Attendance
1	102	Aryan	BSc CS	65	90
2	103	Sahil	BSc IT	88	92

6d. Write a Python program to sort the dataset based on a selected column and display the top 5 records. For example, sort students by marks or products by price.

```
result = df.sort_values("Marks")
print(result)
```

	Roll No	Name	Department	Marks	Attendance
4	105	Kaif	BSc IT	55	70
1	102	Aryan	BSc CS	65	90
6	107	Divyanshu	BSc IT	67	87
3	104	Sumeet	BSc CS	72	80
0	101	Dhruv	BSc IT	78	85
5	106	Usaid	BSc IT	81	75
2	103	Sahil	BSc IT	88	92

6e. Write a Python program to perform data analysis on the dataset by finding the maximum, minimum, average, and total values of a numerical column.

```
print("Maximum Marks ",df["Marks"].max())
print("Minimum Marks ",df["Marks"].min())
print("Average Marks ",df["Marks"].mean())
print("Total Marks ",df["Marks"].sum())
```

Maximum Marks	88
Minimum Marks	55
Average Marks	72.28571428571429
Total Marks	506